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**IntelliCourse: AI Course Builder with Adaptive
Learning Intelligence**

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ABSTRACT:

Traditional learning platforms provide predefined courses and fixed learning paths, while conventional AI chatbots mainly provide one-time responses. Such approaches may not adequately consider prerequisite relationships, learner performance, learning pace, or knowledge retention, resulting in content that may not be appropriately sequenced or personalized.

This project proposes IntelliCourse, an AI-powered Course Builder with Adaptive Learning Intelligence that generates a structured and personalized learning experience from a single user-provided topic. Using a Large Language Model through the Google Gemini API, the system generates structured courses, detailed lessons, relevant YouTube resources, quizzes, coding challenges for programming-related topics, and an AI-powered doubt-solving assistant. Intelligent progress tracking monitors learner performance and supports continuous learning. IntelliCourse introduces three adaptive components: the AI Knowledge Dependency Graph (AKDG), which identifies prerequisite concepts and determines learning sequences; the Learner Digital Twin (LDT), which models learner characteristics including strengths, weaknesses, learning pace, quiz performance, coding performance, and preferred difficulty; and the Adaptive Memory Retention Engine (AMRE), which estimates concept retention and recommends timely revision quizzes. Together with prompt engineering and the Gemini-based AI layer, these components provide context-aware personalization. The project aims to develop a practical and scalable platform that combines Generative AI, learner modeling, knowledge dependency analysis, and adaptive learning.

Keywords: Generative AI, Large Language Model, Adaptive Learning, Personalized Learning, AI Course Generation, AI Knowledge Dependency Graph, Learner Digital Twin, Adaptive Memory Retention Engine, AI Doubt Solver.

[1.]INTRODUCTION:

The rapid growth of Artificial Intelligence (AI) and Generative AI has created new opportunities for improving digital education and personalized learning. Traditional learning platforms generally rely on predefined courses and fixed learning sequences, where the same content and progression are presented to learners regardless of their individual knowledge,

performance, or learning pace. Although such platforms provide structured educational material, they may not adequately address the different learning requirements of individual students. Therefore, intelligent and adaptive learning systems have become an important area of research for providing more personalized and effective educational experiences.

Large Language Models (LLMs) have recently demonstrated significant potential in education by generating explanations, learning content, questions, examples, and conversational responses. AI-powered chatbots can assist learners by answering questions and explaining concepts; however, conventional chatbot-based systems generally operate as question-answering tools and may provide isolated responses without constructing a complete learning pathway. They may also lack awareness of prerequisite relationships between concepts, the learner's previous performance, learning pace, and the likelihood of retaining previously studied concepts. These limitations create a need for learning platforms that can combine Generative AI with learner-specific information and adaptive learning strategies.

An AI-generated course can provide a more structured learning experience by automatically organizing educational content around a user-provided topic. Instead of requiring learners to manually search for courses, lessons, videos, and practice questions, an intelligent course-building system can generate these resources as part of a unified learning pathway. Such a system can generate structured course information, detailed lessons, relevant YouTube learning resources, quizzes, and coding challenges for programming-related topics. In addition, an AI-powered doubt-solving assistant can provide contextual explanations by considering the lesson currently being studied by the learner.

However, simply generating educational content using an LLM does not necessarily make a learning system adaptive. Learners may have different levels of prior knowledge, strengths, weaknesses, learning speeds, and performance patterns. Furthermore, concepts in many academic and technical subjects have prerequisite relationships. For example, a learner may need to understand Python before effectively learning NumPy, followed by Pandas and then Machine Learning. A fixed course sequence may not always represent the most appropriate learning path for every learner. Therefore, identifying concept dependencies and considering individual learner characteristics are important for developing a genuinely personalized learning experience.

To address these limitations, the proposed IntelliCourse system introduces an AI Knowledge Dependency Graph (AKDG). AKDG identifies prerequisite concepts and represents relationships between concepts in a dependency graph. This enables the system to determine an appropriate learning sequence rather than relying only on a fixed syllabus. By considering

prerequisite relationships, IntelliCourse can help learners build foundational knowledge before progressing toward more advanced concepts.

The proposed system also incorporates a Learner Digital Twin (LDT) to maintain a continuously evolving representation of the learner. The LDT models learner-related information such as strengths, weaknesses, learning pace, quiz accuracy, coding performance, preferred difficulty, and learning history. This information can be used by the adaptive learning system to make future learning decisions and provide content that is more appropriate for the individual learner.

Another important challenge in personalized learning is maintaining knowledge over time. A learner may successfully complete a lesson or quiz but gradually forget the concept if it is not reinforced. To address this problem, IntelliCourse introduces an Adaptive Memory Retention Engine (AMRE). AMRE estimates concept retention over time and identifies when reinforcement may be required. Instead of providing separate revision notes, the system can recommend revision quizzes at appropriate times to strengthen long-term knowledge retention. The IntelliCourse architecture combines these adaptive learning components with a Large Language Model through the Google Gemini API. Prompt engineering and relevant learner and lesson context are used to guide the AI in generating structured educational content and contextual responses. The system is implemented using a React frontend, Flask-based microservices, an API Gateway, PostgreSQL, and SQLAlchemy, with the different services responsible for authentication, course management, lesson management, AI functionality, and progress tracking.

The proposed system therefore goes beyond conventional static course platforms and basic AI chatbots by combining Generative AI, knowledge dependency analysis, learner modeling, memory-retention estimation, and progress tracking within a single learning platform. The objective is not to train or fine-tune a new Large Language Model, but to develop an intelligent application around an existing foundation model and use adaptive components to make its educational responses and learning recommendations more personalized.

Overall, IntelliCourse aims to provide a structured, personalized, and continuously adaptive learning experience from a single user-provided topic. The project demonstrates the practical application of Generative AI and adaptive learning intelligence in modern education while remaining suitable for implementation as an MCA Final Year project.

[2.] PROBLEM STATEMENT:

Existing AI-powered learning platforms and conventional AI chatbots can generate educational content and answer learner queries, but they often provide generalized or isolated responses without considering the learner's prior knowledge, concept dependencies, learning performance, and knowledge retention. Fixed learning paths may also require learners to follow the same sequence regardless of their individual strengths, weaknesses, learning pace, and understanding of prerequisite concepts.

Furthermore, simply generating courses and lessons using a Large Language Model does not by itself provide continuous personalization or adaptive learning. There is a need for an integrated system that can identify prerequisite relationships between concepts, maintain a dynamic learner profile, track learning progress, and provide timely reinforcement based on estimated knowledge retention.

Therefore, this project proposes IntelliCourse, an AI-powered Course Builder with Adaptive Learning Intelligence that combines Generative AI with AI Knowledge Dependency Graph (AKDG), Learner Digital Twin (LDT), and Adaptive Memory Retention Engine (AMRE). The system aims to generate structured and personalized learning pathways from a single user-provided topic while providing lessons, learning resources, quizzes, coding challenges for programming-related topics, AI-powered doubt solving, and adaptive learning recommendations.

[3.]OBJECTIVES:

- To develop an AI-powered course generation system that automatically creates a structured learning pathway from a single user-provided topic using the Google Gemini API.
- To generate personalized learning content including structured courses, detailed lessons, learning outcomes, relevant YouTube resources, quizzes, and coding challenges for programming-related topics.
- To develop an AI Knowledge Dependency Graph (AKDG) that identifies prerequisite relationships between concepts and helps generate an appropriate learning sequence.
- To develop a Learner Digital Twin (LDT) that continuously models learner characteristics such as strengths, weaknesses, learning pace, quiz accuracy, coding performance, preferred difficulty, and learning history.
- To develop an Adaptive Memory Retention Engine (AMRE) that estimates concept

retention over time and recommends timely revision quizzes for reinforcing long-term knowledge.

- To implement an AI-powered doubt-solving assistant that uses the current lesson context to provide relevant and personalized explanations to learners.
- To track learner progress and performance through completed lessons, quiz scores, current learning position, and overall course progress.
- To integrate Generative AI with adaptive learning intelligence to provide a personalized and continuously evolving learning experience rather than only generating static educational content.
- To develop the proposed system using a modular microservices architecture with React, Flask, PostgreSQL, SQLAlchemy, an API Gateway, Google Gemini API, and YouTube Data API while keeping the implementation suitable for an MCA Final Year project.

[4.]SCOPE OF THE PROJECT:

The scope of this project is to develop IntelliCourse, an AI-powered Course Builder with Adaptive Learning Intelligence that generates a structured and personalized learning experience from a single user-provided topic. The system will use the Google Gemini API as the Large Language Model to generate educational content and support intelligent learner interactions.

The project will focus on automatically generating structured courses containing course descriptions, learning outcomes, lessons, relevant YouTube learning resources, quizzes, and coding challenges for programming-related topics. An AI-powered doubt-solving assistant will also be provided to help learners understand concepts using the context of the lesson being studied.

A major scope of the project is the development of three adaptive learning components. The AI Knowledge Dependency Graph (AKDG) will identify prerequisite relationships between concepts and support the generation of an appropriate learning sequence. The Learner Digital Twin (LDT) will maintain learner-related information such as strengths, weaknesses, learning pace, quiz accuracy, coding performance, preferred difficulty, and learning history. The Adaptive Memory Retention Engine (AMRE) will estimate concept retention over time and recommend timely revision quizzes to reinforce previously learned concepts.

The project will also include progress tracking to monitor completed lessons, quiz scores, current learning position, and overall course progress. The collected learner performance information will support the adaptive learning components in making future personalization decisions.

The system will be implemented using a React frontend, Flask-based microservices, API Gateway, PostgreSQL, SQLAlchemy, Google Gemini API, and YouTube Data API. The microservices architecture will separate major responsibilities such as authentication, course management, lesson management, AI functionality, and progress tracking while keeping the overall implementation suitable for a single MCA student.

The project will focus on AI-assisted personalized and adaptive learning and will not include the previously excluded features such as teacher dashboards, certificate generation, voice assistants, gamification, multi-language support, final assessments, revision notes, or RAG-based implementation.

[5.] PROPOSED SYSTEM:

The proposed IntelliCourse system aims to develop an AI-powered Course Builder with Adaptive Learning Intelligence that generates a structured and personalized learning experience from a single user-provided topic. The system combines Generative AI, knowledge dependency analysis, learner modeling, memory-retention estimation, and progress tracking to move beyond conventional fixed learning paths and one-time AI responses.

The proposed system follows the general pipeline:

User Topic → Course Generation → Knowledge Dependency Analysis → Course/Lesson Generation → Learning Resources → Quizzes & Coding Challenges → Learner Interaction → Progress Tracking → Learner Digital Twin → Memory Retention Analysis → Adaptive Learning

Initially, the learner provides a topic of interest through the React-based frontend. The request is routed through the API Gateway to the appropriate backend microservice. The Course Service, with support from the AI Service and Google Gemini API, generates a structured course containing the course title, description, difficulty, estimated duration, learning outcomes, course structure, and lesson list.

The generated course is then supported by the AI Knowledge Dependency Graph (AKDG), which identifies prerequisite relationships between concepts and helps determine an appropriate learning sequence. This allows the system to consider concept dependencies rather than relying only on a fixed sequence.

For each lesson, the system generates structured educational content including objectives, explanations, real-world examples, and summaries. Relevant YouTube learning resources are searched through the YouTube Data API and associated with the appropriate lessons. The system also generates MCQ-based quizzes for lessons and coding challenges when the system identifies that the course is programming-related.

During learning, students can interact with the AI-powered doubt-solving assistant. The AI Service receives the learner's question together with the relevant lesson context and uses the Google Gemini API to generate a contextual response. The conversation history is stored for future reference.

The system continuously tracks learner activity through the Progress Service, including completed lessons, quiz performance, current lesson, and overall course progress. This information contributes to the Learner Digital Twin (LDT), which maintains an evolving representation of the learner's strengths, weaknesses, learning pace, quiz accuracy, coding performance, preferred difficulty, and learning history.

The Adaptive Memory Retention Engine (AMRE) uses the learner's learning and performance information to estimate the retention of previously studied concepts over time. When reinforcement is required, the system can recommend an appropriate revision quiz to strengthen knowledge retention. No separate revision-note module is included.

The major system components communicate through the finalized microservices architecture consisting of the API Gateway, Auth Service, Course Service, Lesson Service, AI Service, and Progress Service, with PostgreSQL used for persistent storage. The Google Gemini API provides the underlying generative AI capabilities, while AKDG, LDT, and AMRE provide the adaptive learning intelligence around the AI layer.

The overall proposed system can therefore be represented as:

User Topic → React Frontend → API Gateway → Course/Lesson/AI Services → Gemini API → Generated Learning Content → AKDG → Personalized Learning Path → Learner Interaction → Progress Tracking → LDT → AMRE → Adaptive Learning Recommendations

The proposed system will be evaluated based on its ability to generate structured learning content, provide contextual AI assistance, track learner progress, and support personalized learning through AKDG, LDT, and AMRE. The implementation will remain focused on the finalized scope of IntelliCourse and will not introduce additional modules or technologies beyond the approved project design.

[6.]LITERATURE SURVEY:

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S.NO	TITLE	MERITS	DEMERITS
1	Bringing Generative AI to Adaptive Learning in Education	The study explores the use of Generative AI for adaptive learning and demonstrates the potential of LLM-based approaches to support personalized educational experiences. It provides a foundation for integrating generative AI with adaptive learning strategies	The work primarily focuses on the potential of Generative AI for adaptive learning and does not provide a complete implementation combining course generation, learner modeling, knowledge dependency analysis, and memory retention within a single platform.
2	GPTutor: Great Personalized Tutor with Large Language Models for Personalized Learning Content Generation	The study demonstrates how Large Language Models can be used to generate personalized learning content and provide tutoring support. It highlights the potential of LLMs to adapt educational content according to learner requirements.	Personalized content generation alone may not fully address prerequisite concept relationships, continuous learner modeling, and long-term knowledge retention.
3	How Good is ChatGPT in Giving Adaptive Guidance Using Knowledge Graphs in E-	The study investigates the combination of LLM-based guidance with knowledge graphs for adaptive learning. It highlights the usefulness of structured knowledge relationships for	The work focuses mainly on adaptive guidance using knowledge graphs and does not provide the complete course-generation, learner digital twin, and memory-retention

	Learning Environments?	providing more context-aware educational guidance.	framework proposed in IntelliCourse.
4	A Knowledge Graph Enabled Intelligent Adaptive Learning Mode	The research demonstrates how knowledge graphs can support intelligent adaptive learning by representing relationships between concepts and supporting personalized learning paths. This provides a strong foundation for the AI Knowledge Dependency Graph (AKDG) component of IntelliCourse.	Knowledge-graph-based adaptive learning alone does not provide the complete Generative AI-based course generation, AI doubt solving, learner digital twin, and adaptive memory-retention functionality required by the proposed system.
5	How Teachers Can Use Large Language Models and Bloom's Taxonomy to Create Educational Quizzes	The study demonstrates the usefulness of LLMs for generating educational quizzes and highlights the potential of structured approaches for producing questions at different cognitive levels.	The study focuses primarily on quiz generation and does not address complete course generation, learner modeling, concept dependencies, or adaptive memory retention.
6	Personalized Cloze Test Generation with Large Language Models	The research demonstrates the ability of LLMs to generate personalized assessment content. It supports the use of Generative AI for creating learner-specific practice material.	The work focuses on cloze-test generation and does not cover a complete adaptive learning platform with courses, lessons, quizzes, coding challenges, learner modeling, and concept dependency analysis.
7	Enhancing Computer Programming Education with Large Language Models: A Study on Effective Prompt Engineering for Python Code Generation	The study highlights the importance of prompt engineering when using LLMs for programming education and code generation. This supports the use of structured prompts for IntelliCourse's programming-related coding challenges and AI assistance.	The study focuses on programming and Python code generation rather than a general-purpose learning platform. It does not address adaptive learning pathways, learner digital twins, or memory retention.
8	Embracing the Future of AI in the Classroom: Prompt Engineering and Generative AI in Education	The research discusses the role of prompt engineering and Generative AI in educational environments. It supports the use of carefully designed prompts for obtaining useful and contextually appropriate educational responses from LLMs.	Prompt engineering by itself does not provide learner adaptation, concept dependency management, progress modeling, or long-term memory-retention mechanisms.

9.	Enhancing Engineering Education Through LLM-Driven Adaptive Quiz Generation	The study demonstrates the potential of LLMs for generating adaptive quizzes in educational environments. It supports the use of AI-generated assessments for evaluating learner understanding and performance.	The focus is primarily on adaptive quiz generation and does not cover complete AI course construction, prerequisite concept graphs, learner digital twins, or adaptive memory retention.
10	Transforming Engineering Education Using Generative AI and Digital Twin Technologies	The study explores the combination of Generative AI and digital twin technologies in education. It provides conceptual support for representing learners digitally and using AI to improve personalized educational experiences.	The study does not provide the complete combination of course generation, knowledge dependency analysis, learner modeling, memory retention, quizzes, coding challenges, and AI doubt solving implemented in IntelliCourse.
11	Applications of Large Language Models in Education: Literature Review and Case Study	The study reviews applications of Large Language Models in education and highlights their potential for individualized practice, immediate feedback, and personalized educational support. It provides a strong foundation for using LLMs as an educational assistant in IntelliCourse.	The study primarily discusses the applications and potential of LLMs rather than providing an integrated adaptive learning system combining course generation, learner modeling, knowledge dependencies, and retention-based learning.
12	A Systematic Review of Knowledge Tracing and Large Language Models in Education: Opportunities, Issues, and Future Research	The study examines the relationship between Knowledge Tracing and LLMs for estimating learner knowledge states and supporting personalized learning. It is particularly relevant to the learner-modeling aspect of IntelliCourse.	Existing Knowledge Tracing approaches can face challenges related to interpretability, data requirements, scalability, and generalization. The study does not provide the complete AKDG, LDT, and AMRE framework proposed in IntelliCourse.
13	Generative AI and Its Impact on Personalized Intelligent Tutoring Systems	The study explores the integration of Generative AI and LLMs with Intelligent Tutoring Systems for personalized education. It discusses dynamic content generation, adaptive learning pathways, automated question generation, personalized feedback, and interactive dialogue systems, which are closely related to IntelliCourse.	The study identifies challenges including pedagogical accuracy, bias, learner engagement, and reliability of AI-generated educational content. It does not implement the complete IntelliCourse architecture with AKDG, LDT, and AMRE.

14	From Prediction to Application: Language Model-based Code Knowledge Tracing with Domain Adaptive Pre-Training and Automatic Feedback System with Pedagogical Prompting for Comprehensive Programming Education	The study applies language-model-based Knowledge Tracing to programming education and combines learner-performance analysis with personalized feedback. It is relevant to IntelliCourse's programming-related coding challenges and learner-performance modeling.	The approach is primarily focused on programming education and code knowledge tracing rather than general-purpose course generation across multiple learning domains.
15	AI-Driven Personalized Learning Systems: Enhancing Educational Effectiveness	The study examines AI-driven personalized learning using techniques such as machine learning, natural language processing, and knowledge representation. It supports the use of AI and learner information to create adaptive learning experiences.	The work provides a broad perspective on AI-driven personalized learning rather than implementing an integrated Generative AI platform containing course generation, concept dependency analysis, learner digital twin, and adaptive memory retention.

[6.1] FINDINGS IN LITERATURE SURVEY:

The literature survey indicates that Generative AI and Large Language Models (LLMs) have significant potential to improve personalized and adaptive learning environments. The major findings from the reviewed studies are summarized below:

1. Generative AI can support personalized learning:

Recent studies demonstrate that LLMs can generate educational content, explanations, tutoring responses, and learning materials according to learner requirements.

2. LLMs can automate educational content generation:

Research shows that LLMs can effectively generate learning materials such as lessons, quizzes, questions, explanations, and programming-related content, reducing the need for completely manual content creation.

3. Knowledge graphs can improve learning-path personalization:
Knowledge-graph-based approaches demonstrate that representing relationships between concepts and prerequisites can help generate more meaningful learning sequences instead of relying only on fixed learning paths.
4. Prompt engineering is important for educational applications:
The reviewed studies indicate that carefully designed prompts can improve the relevance, structure, and usefulness of LLM-generated educational responses and programming content.
5. AI-generated quizzes can support learner assessment:
LLM-based quiz generation can provide learners with automatically generated practice and assessment material and can contribute to evaluating learner understanding.
6. Digital twin technology can support learner modeling:
The literature indicates that digital twin concepts can be used to represent learner characteristics and support more personalized educational experiences.
7. Existing approaches generally focus on individual capabilities:
Most reviewed studies concentrate on specific areas such as personalized content generation, quiz generation, knowledge graphs, programming assistance, or digital twins rather than integrating all these capabilities into one learning platform.
8. Long-term knowledge retention requires additional adaptive mechanisms:
The reviewed literature highlights personalization and adaptive learning, but there is scope for integrating a dedicated mechanism that estimates concept retention over time and recommends timely reinforcement.
9. A gap exists for an integrated Generative AI-based adaptive learning platform:
The literature suggests an opportunity to combine Generative AI, knowledge dependency analysis, learner modeling, progress tracking, and memory-retention estimation into a unified learning system.
10. Research gap addressed by IntelliCourse:
Based on these findings, IntelliCourse proposes an integrated framework combining AI Knowledge Dependency Graph (AKDG), Learner Digital Twin (LDT), and Adaptive

Memory Retention Engine (AMRE) with AI-generated courses, lessons, quizzes, coding challenges, YouTube resources, progress tracking, and an AI-powered doubt-solving assistant.

[7.] METHEDOLOGY:

The proposed methodology for IntelliCourse focuses on developing an AI-powered Course Builder with Adaptive Learning Intelligence that generates structured learning content from a single user-provided topic and continuously adapts the learning experience based on concept dependencies, learner performance, and estimated knowledge retention.

Step 1: User Topic Input

The learner provides a topic of interest through the React-based frontend. The topic may be related to programming, computer science, mathematics, or another learning domain.

The request is sent through the API Gateway to the appropriate backend service.

Step 2: AI Course Generation

The provided topic is processed by the Course Service with the support of the AI Service and Google Gemini API.

The system generates:

- Course title
- Description
- Difficulty level
- Estimated duration
- Learning outcomes
- Course structure
- Lesson list

The generated course is stored in the PostgreSQL database.

Step 3: AI Knowledge Dependency Graph (AKDG)

The generated topic and concepts are analyzed to identify prerequisite relationships between concepts.

The AI Knowledge Dependency Graph (AKDG) represents these relationships and helps determine an appropriate learning sequence.

For example:

Python



NumPy



Pandas



Machine Learning



Deep Learning

This allows IntelliCourse to consider concept dependencies instead of relying only on a fixed learning sequence.

Step 4: AI Lesson Generation

Based on the generated course structure and learning sequence, the Lesson Service generates individual lessons using the Gemini API.

Each lesson contains:

- Lesson title
- Learning objectives
- Explanation
- Real-world examples
- Summary

The generated lesson information is stored in PostgreSQL.

Step 5: YouTube Learning Resource Integration

Relevant YouTube videos are searched for each lesson using the YouTube Data API.

The system associates appropriate video resources with the corresponding lessons.

Only YouTube video embedding is used; videos are not downloaded.

Step 6: Quiz and Coding Challenge Generation

The AI Service generates MCQ-based quizzes for the lessons.

Each quiz contains:

- Question
- Options
- Correct answer
- Explanation

The system also determines whether the generated course is programming-related.

For programming-related courses, coding challenges are generated. For non-programming topics, coding challenges are not generated.

Step 7: AI Doubt-Solving Assistant

The learner can ask questions related to the lesson currently being studied.

The AI Service receives:

- Learner's question
- Relevant lesson context
- Available learner context where applicable

The information is provided to the Gemini API through structured prompts so that the generated response is relevant to the learner's current lesson.

The interaction is stored in AI Chat History.

Step 8: Learner Progress Tracking

The system continuously records learner activity through the Progress Service.

The tracked information includes:

- Completed lessons
- Quiz scores
- Current lesson
- Course progress

This information is stored in PostgreSQL and is used by the adaptive learning components.

Step 9: Learner Digital Twin (LDT)

The learner's performance and learning history are used to maintain the Learner Digital Twin (LDT).

The LDT dynamically models:

- Strong concepts
- Weak concepts
- Learning speed
- Quiz accuracy
- Coding performance
- Preferred difficulty
- Learning history

The learner model can subsequently be used to support personalized learning decisions.

Step 10: Adaptive Memory Retention Engine (AMRE)

The Adaptive Memory Retention Engine (AMRE) analyzes learning and performance information to estimate the retention of previously studied concepts over time.

When a concept requires reinforcement, AMRE recommends a suitable revision quiz.

The system therefore provides reinforcement through quizzes rather than introducing a separate revision-notes module.

Step 11: Adaptive Learning

The information generated by AKDG, LDT, and AMRE is used together with the learner's current progress to support personalized learning decisions.

The overall adaptive process can be represented as:

AKDG



Concept Dependencies



Learning Sequence



Learner Activity



LDT



Learner Performance & Profile



AMRE



Retention Estimation



Revision Quiz Recommendation



Adaptive Learning Experience



Step 12: System Integration and Evaluation

All major components are integrated through the finalized microservices architecture consisting of:

- API Gateway
- Auth Service
- Course Service
- Lesson Service
- AI Service
- Progress Service

The system uses:

- React for the frontend
- Flask for backend services
- PostgreSQL for persistent storage
- SQLAlchemy for database interaction
- Google Gemini API for Generative AI
- YouTube Data API for learning resources

The completed system will be evaluated based on its ability to:

- Generate structured courses
- Generate appropriate lessons and learning resources
- Generate quizzes and programming challenges where applicable
- Provide contextual AI doubt solving
- Track learner progress

- Maintain learner-specific information through LDT
- Identify concept dependencies through AKDG
- Provide retention-based revision quiz recommendations through AMRE

The methodology therefore integrates Generative AI with knowledge dependency analysis, learner modeling, progress tracking, and adaptive memory retention to create a personalized learning platform while maintaining the finalized scope and architecture of the MCA project.

[8.] SOFTWARE REQUIREMENTS: Functional and Non-Functional Requirements

[8.1] Functional Requirements

The proposed IntelliCourse system shall:

- **User Registration:** Allow new users to create an account by providing the required registration information.
- **User Authentication:** Allow registered users to securely log in and access their learning profile using authenticated access.
- **Profile Management:** Allow users to view and manage their profile information.
- **Topic Input:** Accept a single user-provided learning topic through the application interface.
- **AI Course Generation:** Generate a structured course containing the course title, description, difficulty, estimated duration, learning outcomes, course structure, and lesson list using the Google Gemini API.
- **AI Knowledge Dependency Analysis:** Identify prerequisite concepts and establish concept dependencies through the AI Knowledge Dependency Graph (AKDG) to support an appropriate learning sequence.
- **AI Lesson Generation:** Generate lessons containing lesson titles, objectives, explanations, real-world examples, and summaries.
- **YouTube Resource Integration:** Search for and associate relevant YouTube learning videos with lessons using the YouTube Data API.
- **Quiz Generation:** Generate MCQ-based quizzes for lessons containing questions, options, correct answers, and explanations.
- **Coding Challenge Generation:** Detect programming-related courses and generate coding challenges for them. Non-programming courses shall not generate coding challenges.
- **Progress Tracking:** Track completed lessons, quiz scores, current lesson, and overall course progress.
- **AI Doubt Solving:** Allow learners to ask questions related to the current lesson and provide contextual AI-generated responses using the lesson context.
- **AI Chat History:** Store the learner's questions and AI responses as part of the AI chat history.

- **Learner Digital Twin:** Maintain a continuously evolving learner model containing information such as strengths, weaknesses, learning speed, quiz accuracy, coding performance, preferred difficulty, and learning history.
- **Adaptive Memory Retention:** Estimate concept retention over time using the Adaptive Memory Retention Engine (AMRE) and recommend timely revision quizzes when reinforcement is required.
- **Adaptive Learning:** Use information from AKDG, LDT, AMRE, and learner progress to support personalized learning decisions.
- **Database Management:** Store and retrieve user, course, lesson, video, quiz, coding challenge, progress, chat history, and learner digital twin information using PostgreSQL.

[8.2] Non-Functional Requirements

Performance

The system should generate course and lesson content, process learner interactions, and provide AI responses within a reasonable response time under normal operating conditions.

Efficiency

The system should use a modular microservices architecture so that individual services can be developed, maintained, and scaled independently when required.

Reliability

The system should maintain consistent behavior and correctly store and retrieve learner, course, and progress information under normal operating conditions.

Security

The system should protect user information and authentication credentials. Passwords should be securely encrypted, and authenticated APIs should prevent unauthorized access to user-specific information.

Maintainability

The system should use a modular architecture with separate services for authentication, course management, lesson management, AI functionality, and progress tracking. The AKDG, LDT, and AMRE components should also remain logically modular.

Usability

The system should provide a clear and user-friendly interface through which learners can enter topics, access generated courses and lessons, attempt quizzes, view learning resources, track progress, and interact with the AI doubt-solving assistant.

Data Integrity

The system should maintain accurate and consistent relationships between users, courses, lessons, quizzes, coding challenges, videos, progress records, chat history, and learner digital twin information.

AI Response Quality

The system should use structured prompt engineering and appropriate lesson and learner context to generate relevant and coherent educational responses through the Gemini API.

[9.] SYSTEM ARCHITECTURE:

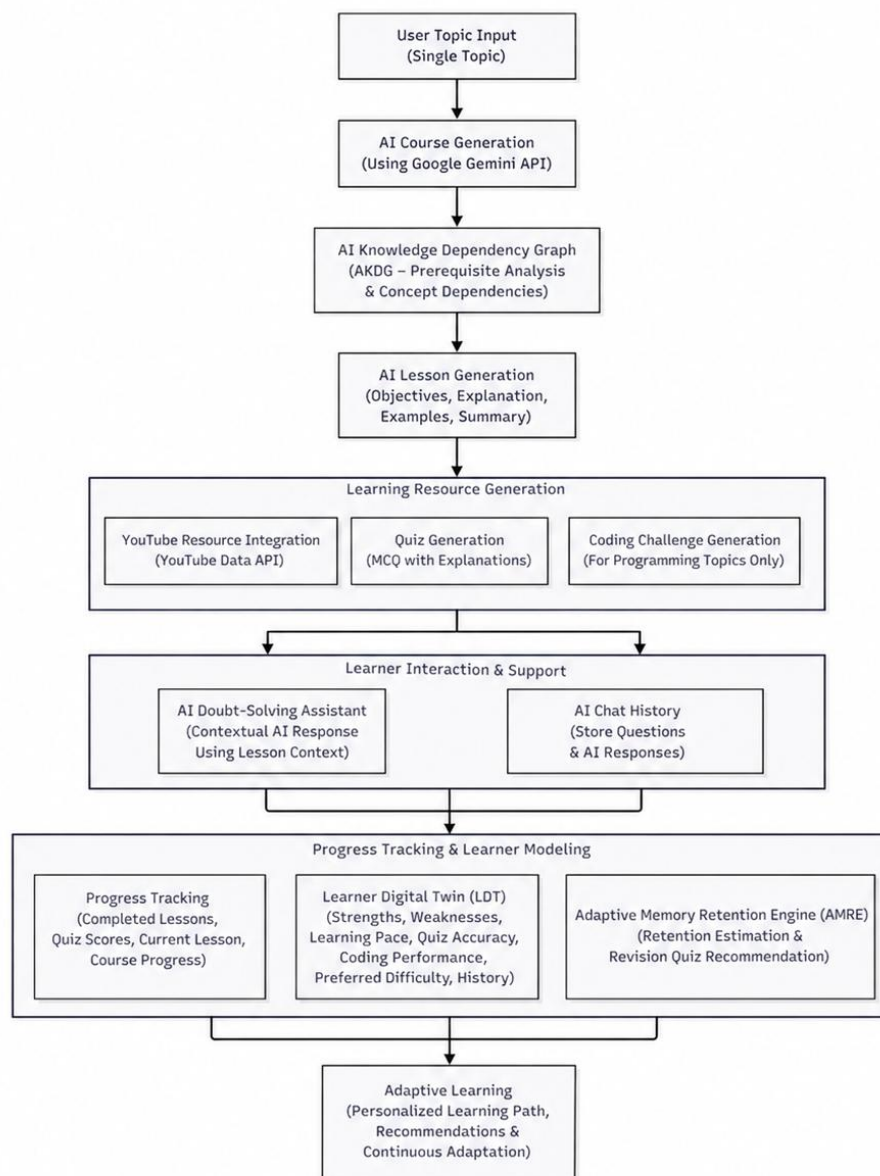


Fig. 9.1 System Architecture

[10]. UML DIAGRAMS:

Use Case Diagram

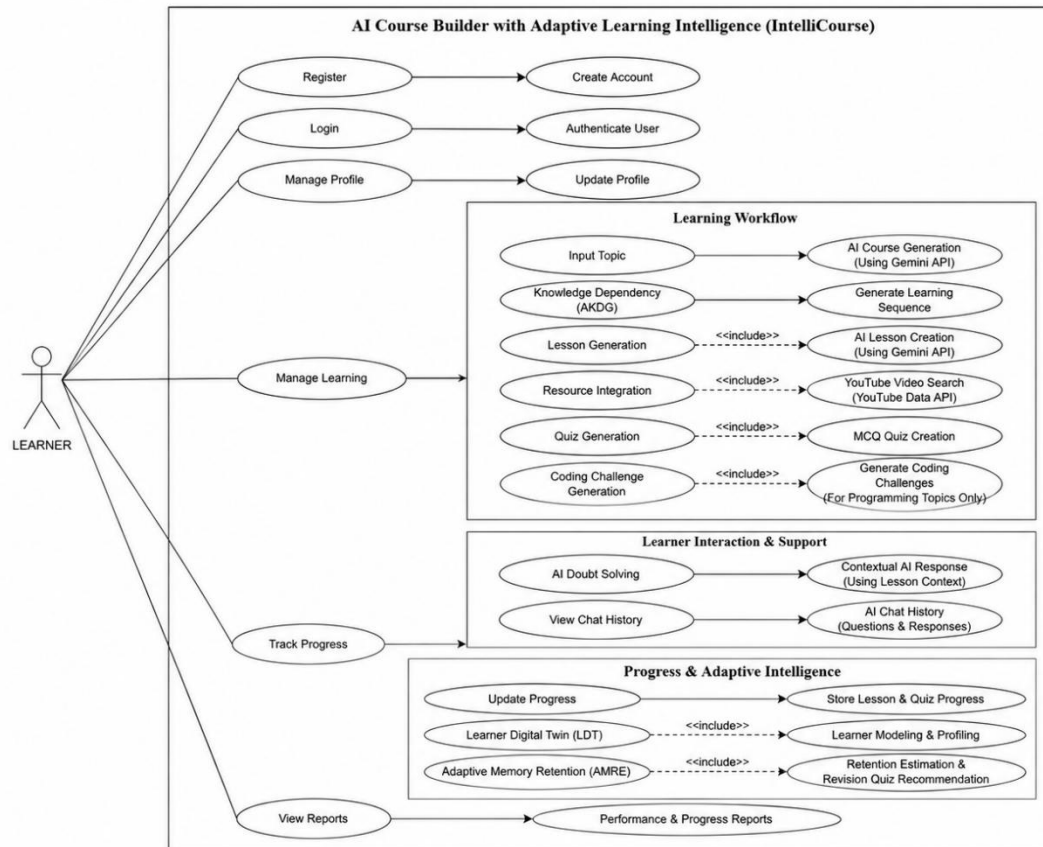


Fig. 10.1 Use Case Diagram

Class Diagram

Class Diagram

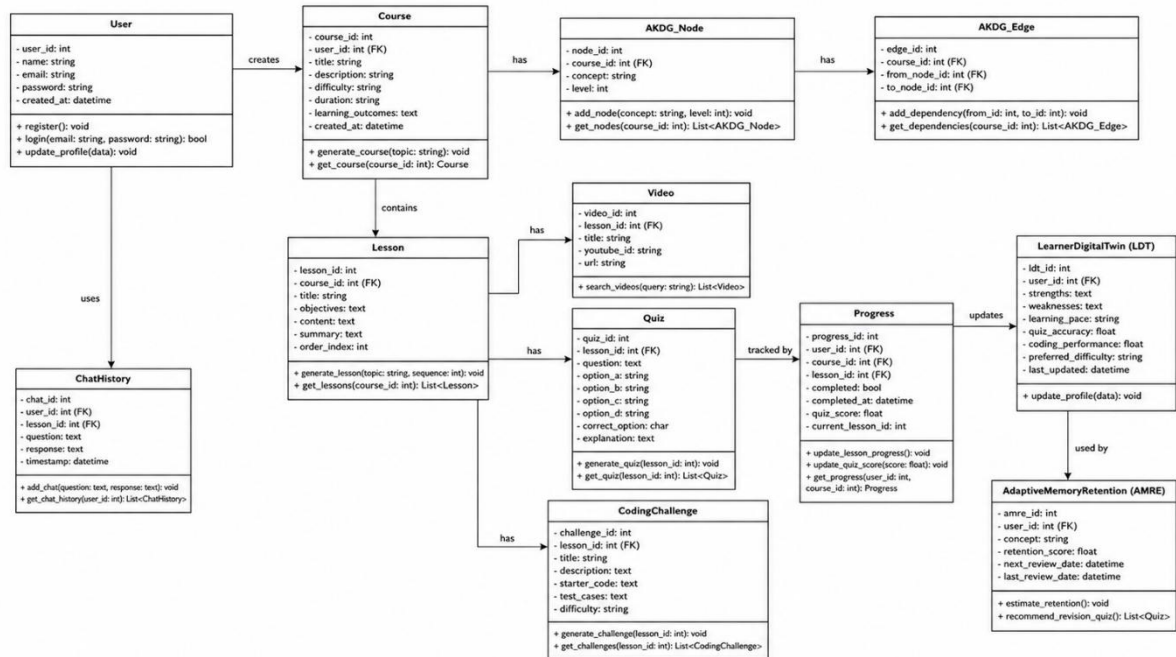


Fig. 10.2 Class Diagram

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